

Supplementary Information for: “Observation of continuous time crystals and quasi-crystals in spin gases”

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I. EXPERIMENTAL DETAILS

In this section, we provide experimental details about our spin system, spin-exchange optical pumping, the optical detection setup, and the feedback configuration.

A. Spin system

A 0.5 cm³ vapor cell contains 50 Torr of ¹²⁹Xe (corresponding to a number density of $1.8 \times 10^{24} \text{ m}^{-3}$), a droplet of ⁸⁷Rb providing saturated vapor, and 250 Torr N₂ as buffer gas. The cell is heated to 105°C, yielding a saturated ⁸⁷Rb vapor number density of $8.3 \times 10^{18} \text{ m}^{-3}$. The ⁸⁷Rb atoms are optically polarized through a circularly polarized pump beam with a power of approximately 11.5 mW, tuned to the D1 transition (795 nm) of ⁸⁷Rb. Noble gas lacks any optical transition from the ground levels, making them difficult to polarize directly using optical pumping methods. Instead, noble gases can be polarized through spin exchange when placed in the same vapor cell as optically pumped alkali-metal atoms.

B. Spin exchange optical pumping

The primary exchange processes involve binary collisions and the formation and destruction of van der Waals molecules via three-body collisions [1, 2]. At low Xe gas pressures (well below one atmosphere), spin-exchange interactions are predominantly governed by three-body collisions. In the three-body collision scenario, an Rb atom and a Xe atom collide in the presence of a third body, in this case, a N₂ molecule or a Xe atom, forming a weakly bound van der Waals molecule. This molecule then evolves freely for a duration of τ_{vdw} before being disrupted by a subsequent collision. During the relatively long molecular lifetime, $\tau_{\text{vdw}} \approx 10^{-9} \text{ s}$, the weak coupling between the electron spin **S** of the alkali-metal atom, the rotational angular momentum **N** of the molecule, and the nuclear spin **K** of the noble-gas atom leads to the flipping down of **S** and partial flipping up of **K**. Consequently, a fraction of the alkali metal spin polarization is transferred to the noble-gas nucleus. In binary collisions, as the lifetime of the binary collision, τ_{bc} , is very short ($\approx 10^{-12} \text{ s}$), only the nuclear spin **K** and the electron spin **S** couple, also resulting in the transfer of electron spin polarization to the nuclear spin.

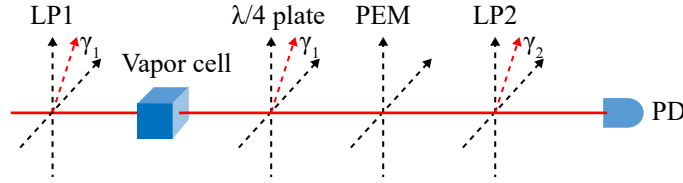
C. Optical probe

The spin polarization of ¹²⁹Xe is measured via the electron spin polarization **P^e** of ⁸⁷Rb atoms through spin exchange interaction. The transverse electron spin polarization of ⁸⁷Rb along *x* is measured through the optical rotation of the probe beam tuned to the D2 transition of ⁸⁷Rb [3, 4]:

$$\theta = \frac{1}{4} l r_e c f_{\text{D2}} [\text{Rb}] D(\nu_{\text{pr}}) P_x^e, \quad (\text{S1})$$

where $l \approx 0.8 \text{ cm}$ is the optical path length, $r_e = 2.8 \times 10^{-13} \text{ cm}$ is the classical radius of the electron, c is the speed of light, $f_{\text{D2}} \approx 2/3$ is the oscillator strength, $[\text{Rb}]$ is the number density of ⁸⁷Rb atoms, $D(\nu_{\text{pr}}) = (\nu_{\text{pr}} - \nu_{\text{D2}}) / [(\nu_{\text{pr}} - \nu_{\text{D2}})^2 + (\Delta\nu_{\text{D2}}/2)^2]$, ν_{pr} is the frequency of the probe laser light, $\Delta\nu_{\text{D2}}$ is the full-width at half-maximum (FWHM) of the optical D2 transition of frequency ν_{D2} . In this experiment, the probe beam power is approximately 2 mW.

The Faraday rotation angle is typically very small and requires a precise detection setup. Considering the low-frequency noise (covering DC-100 Hz) in experiment, such as 1/*f* noise from photodetectors, airflow, mechanical vibrations of optical components, and fluctuations in laser power and wavelength, a photoelastic modulator (PEM) is used to shift the low-frequency Faraday rotation angle signal into a higher frequency range (specifically, 50 kHz in this case). The PEM is a phase modulator based on the photoelastic effect in a crystal, where the phase difference between its fast and slow axes varies periodically under cyclic mechanical pressure: $A(t) = A_0 \sin \Omega_m t$, where A_0 is the modulation amplitude, typically near a few radians (e.g., 2.405 rad to optimize 1st-order Bessel function), Ω_m is the modulation frequency (e.g., 50 kHz). As the probe light passes through the crystal, its phase undergoes periodic modulation, shifting of the originally low-frequency optical signal into a higher frequency range. The experimental optical path is shown in Fig. 1. A linearly polarized probe beam passes through a first polarizer, an atomic vapor cell, a $\lambda/4$ wave plate, a PEM and a second polarizer in sequence, and finally enters the photodiode. The optical axis of the PEM is aligned with the *x*-axis. The fast axis of the quarter-wave plate is aligned parallel to the transmission axis of the first polarizer (LP1). Assuming that the angles between the polarization axes of the first and second polarizers and the *x*-axis are denoted by γ_1 and γ_2 , respectively. In the following, we calculate the probe light intensity after it traverses the optical path depicted in Fig. 1 using Stokes vectors and the Mueller matrix.



Supplementary Fig. 1 **Optical path diagram of the photoelastic modulation scheme.** The optical axis of the PEM is aligned with the horizontal axis (x-axis). The fast axis of the quarter-wave ($\lambda/4$) plate is aligned parallel to the transmission axis of the first polarizer (LP1). γ_1 and γ_2 denote the angles between the polarization axes of LP1 and LP2 and the x-axis, respectively.

The Stokes vector of the incident light passing through LP1 is expressed as:

$$\mathbf{S}_1 = I_0 \begin{bmatrix} 1 & \cos 2\gamma_1 & \sin 2\gamma_1 & 0 \end{bmatrix}^T. \quad (\text{S2})$$

After passing through the cell, it transforms to:

$$\mathbf{S}_2 = M_{\text{cell}}(\theta) \cdot \mathbf{S}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta & 0 \\ 0 & -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \\ \sin 2\gamma_1 \\ 0 \end{bmatrix} = I_0 \begin{bmatrix} 1 \\ \cos(2\gamma_1 - 2\theta) \\ \sin(2\gamma_1 - 2\theta) \\ 0 \end{bmatrix}. \quad (\text{S3})$$

Subsequently, passing through a quarter-wave plate modifies it to:

$$\begin{aligned} \mathbf{S}_3 = M_{\lambda/4}(\gamma_1) \cdot \mathbf{S}_2 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\gamma_1 & \sin 2\gamma_1 \cos 2\gamma_1 & -\sin 2\gamma_1 \\ 0 & \sin 2\gamma_1 \cos 2\gamma_1 & \sin^2 2\gamma_1 & \cos 2\gamma_1 \\ 0 & \sin 2\gamma_1 & -\cos 2\gamma_1 & 0 \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos(2\gamma_1 - 2\theta) \\ \sin(2\gamma_1 - 2\theta) \\ 0 \end{bmatrix} \\ &= I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \sin 2\gamma_1 \cos 2\theta \\ \sin 2\theta \end{bmatrix}. \end{aligned} \quad (\text{S4})$$

Then the PEM modulate it as:

$$\begin{aligned} \mathbf{S}_4 = M_{\text{PEM}}(A(t), 0^\circ) \cdot \mathbf{S}_3 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos A(t) & \sin A(t) \\ 0 & 0 & -\sin A(t) & \cos A(t) \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \sin 2\gamma_1 \cos 2\theta \\ \sin 2\theta \end{bmatrix} \\ &= I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \cos A(t) \sin 2\gamma_1 \cos 2\theta + \sin A(t) \sin 2\theta \\ -\sin A(t) \sin 2\gamma_1 \cos 2\theta + \cos A(t) \sin 2\theta \end{bmatrix}. \end{aligned} \quad (\text{S5})$$

The Mueller matrix for LP2 is:

$$M_{\text{LP}}(\gamma_2) = \frac{1}{2} \begin{bmatrix} 1 & \cos 2\gamma_2 & \sin 2\gamma_2 & 0 \\ \cos 2\gamma_2 & \cos^2 2\gamma_2 & \sin 2\gamma_2 \cos 2\gamma_2 & 0 \\ \sin 2\gamma_2 & \sin 2\gamma_2 \cos 2\gamma_2 & \sin^2 2\gamma_2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{S6})$$

After passing through LP2, the output light intensity is the first component of the Stokes vector after the action of the LP2 Mueller matrix:

$$I_{\text{out}} = \frac{I_0}{2} [1 + \cos 2\gamma_2 \cos 2\gamma_1 \cos 2\theta + \sin 2\gamma_2 (\cos A(t) \sin 2\gamma_1 \cos 2\theta + \sin A(t) \sin 2\theta)] \quad (\text{S7})$$

Since the Faraday rotation angle θ is very small, we apply the small-angle approximations $\cos \theta \approx 1$ and $\sin \theta \approx \theta$, thus :

$$I_{\text{out}} \approx \frac{I_0}{2} [1 + \cos 2\gamma_1 \cos 2\gamma_2 + \sin 2\gamma_1 \sin 2\gamma_2 \cos A(t) + 2\theta \sin 2\gamma_2 \sin A(t)] \quad (\text{S8})$$

The terms $\cos A(t) = \cos(A_0 \sin \Omega_m t)$ and $\sin A(t) = \sin(A_0 \sin \Omega_m t)$ can be expanded into Bessel function series. Given that A_0 is small, expanding up to second order yields the approximations: $\cos A(t) \approx 1 - \frac{A_0^2}{4} + \frac{A_0^2}{4} \cos 2\Omega_m t + O(A_0^4)$, $\sin A(t) \approx A_0 \sin \Omega_m t + O(A_0^3)$. Thus, the output light intensity is approximately:

$$I_{\text{out}} \approx \frac{I_0}{2} [1 + \cos(2\gamma_1 - 2\gamma_2)] + \theta A_0 I_0 \sin 2\gamma_2 \sin \Omega_m t - \frac{A_0^2}{8} I_0 \sin 2\gamma_1 \sin 2\gamma_2 \cos 2\Omega_m t. \quad (\text{S9})$$

From above equations, we can determine that the desired signal is directly proportional to the first-order modulation frequency component.

- To maximize the first harmonic, $\sin 2\gamma_2 = \pm 1$ should be satisfied;
- To minimize the DC signal, $\cos 2(\gamma_1 - \gamma_2) = -1$ should be satisfied.

To maximize the desired signal, we set $\gamma_2 = \pm \pi/4$, $\gamma_1 = \mp \pi/4$

D. Feedback

The probe signal, which contains information about spin polarization, is captured by a photodetector. After demodulation with a lock-in amplifier, the signal is converted into a voltage signal V . This voltage is then fed through a series resistor R to y-direction coils, thereby generating a feedback magnetic field B_f . To control the activation of the feedback, a relay is incorporated into the circuit. The relay switch is controlled by a TTL digital output module (NI 9401), which outputs a digital voltage of either 0 or 5 V. When the TTL signal is high, the relay connects, activating the feedback; when the signal is low, the relay disconnects, deactivating the feedback. For consistent determination of the relative time phase across different experimental runs, a common time origin is necessary. Both the TTL signal and the probe signal are synchronously recorded using a 4-channel, 24-bit simultaneous analog input module (NI 9239). The rising edge of the TTL signal is designated as the temporal zero point. Notably, the NI 9239 is a simultaneous sampling ADC module, ensuring no inter-channel delay.

The key parameters of the feedback system that influence the relative time phase. Critical factors affecting timing accuracy is the rising edge timing of the TTL signal, denoted as τ_{up} . Based on our 20 experimental calibrations, we determined that the rising edge τ_{up} of the TTL signal does not exceed 0.04 ms. This level of timing uncertainty has a negligible impact on the time phase distribution. For example, in comparison to the period of the CTC (approximately 57.1 ms at a frequency of 17.5 Hz), any potential phase shift caused by these uncertainties is at most $0.04/57.1 \times 2\pi \approx 0.0014\pi$. This is far too small to account for the observed 2π random phase distribution.

II. MODEL AND DETAILED DERIVATIONS

The dynamics of noble-gas atoms in our work are governed by the underlying spin Hamiltonian, from which Eq. (1) of the main text can be derived. To provide a clearer explanation, we first identify the relevant spin Hamiltonian and its connection to Eq. (1), and then provide a detailed derivation of each spin Hamiltonian.

The spin Hamiltonian for the noble-gas system is expressed as

$$H = -\mu_K \cdot \mathbf{B}_z - \mu_K \cdot \mathbf{B}_f + \alpha \mathbf{S} \cdot \mathbf{K}, \quad (\text{S10})$$

where the first two terms represent the Zeeman interaction of the nuclear spin with the applied static magnetic field B_z and the the feedback magnetic field $\mathbf{B}_f$, and the last term describes the spin-exchange interaction between the noble-gas nuclear spin $\mathbf{K}$ and the electron spin $\mathbf{S}$ of the alkali-metal atoms. The nuclear magnetic moment is given by $\mu_K = \gamma \mathbf{K}$, where γ denotes the gyromagnetic ratio of noble gas. Each term in Eq. (1) corresponds to a specific contribution from the Hamiltonian. The explicit form of Eq. (1) is provided below, with the related spin Hamiltonian labeled accordingly

$$\left\{ \begin{array}{l} \dot{P}_x(\omega) = \overbrace{\omega P_y(\omega)}^{-\mu_K \cdot \mathbf{B}_z} + \overbrace{\alpha P_z(\omega) \int \rho(\omega') P_x(\omega') d\omega'}^{-\mu_K \cdot \mathbf{B}_f} - \overbrace{\frac{P_x(\omega)}{T_2}}^{\text{relaxation}} \\ \dot{P}_y(\omega) = -\overbrace{\omega P_x(\omega)}^{-\mu_K \cdot \mathbf{B}_z} - \overbrace{\frac{P_y(\omega)}{T_2}}^{\text{relaxation}} \\ \dot{P}_z(\omega) = \overbrace{\alpha P_x(\omega) \int \rho(\omega') P_x(\omega') d\omega'}^{-\mu_K \cdot \mathbf{B}_f} - \overbrace{\frac{P_z(\omega)}{T_1}}^{\text{relaxation}} + \overbrace{R_{\text{SE}}}^{\alpha \mathbf{S} \cdot \mathbf{K}} \end{array} \right.$$

In the following analysis, we present a comprehensive derivation of Eq. (1) of the main text originating from the associated spin Hamiltonian, carefully detailing the approximations applied throughout.

(1) Spin-exchange interactions. The magnetic dipole hyperfine interaction is the primary mechanism responsible for the spin exchange, described by the Hamiltonian: $H_{SE} = \alpha \mathbf{S} \cdot \mathbf{K}$. The coupling coefficient is given by $\alpha = \frac{8\pi g_S \mu_B \gamma}{3} |\psi(0)|^2$, where $|\psi(0)|^2$ represents the probability density of the alkali-metal atom's valence electron at the site of the noble-gas nucleus, g_S denotes the electron's Landé g-factor, and μ_B is the Bohr magneton. By averaging over all types of collisions, the rate of change of the noble gas's density matrix ρ due to spin-exchange interaction is expressed as follows [1, 5]:

$$\begin{aligned} \frac{d}{dt}\rho = & (\Gamma_{bc} + \Gamma_{vdw,S}) \sum_{m,n=x,y,z} 4\langle S_m S_n \rangle \left(K_n \rho K_m - \frac{1}{2} \{K_m K_n, \rho\} \right) \\ & + \Gamma_{vdw,F} \sum_{m,n=x,y,z} 4\langle F_m F_n \rangle \left(K_n \rho K_m - \frac{1}{2} \{K_m K_n, \rho\} \right) + \frac{1}{i} [\mathcal{E}_{SE}, \rho], \end{aligned} \quad (S11)$$

where Γ_{bc} , $\Gamma_{vdw,S}$ and $\Gamma_{vdw,F}$ denote the binary collision rate and the three-body formation rates of molecules with “very short” and “short” lifetimes, respectively; and F represents the total angular momentum of the ^{87}Rb atoms. The frequency-shift operator $\mathcal{E}_{SE} = \frac{8\pi g_S \mu_B \gamma}{3} \kappa[\text{Rb}] \langle \mathbf{S} \rangle \cdot \mathbf{K}$ can be understood as the effect of an effective magnetic field $\mathbf{B}_{SE}$ generated by the spin-polarized alkali-metal atoms with a number density $[\text{Rb}]$. We define the spin polarization as $\mathbf{P} = \langle \mathbf{K} \rangle / K = 2\langle \mathbf{K} \rangle$. By multiplying both sides of the equation by $2\mathbf{K}$ and taking the trace, we find that the evolution of the nuclear spin polarization is given by

$$\begin{aligned} \frac{d}{dt}\mathbf{P} = & (\Gamma_{bc} + \Gamma_{vdw,S})(2\langle \mathbf{S} \rangle - \mathbf{P}) + \Gamma_{vdw,F}(2\langle \mathbf{F} \rangle - \langle 2\mathbf{F} \cdot \mathbf{F} - \mathbf{F}\mathbf{F} \cdot - (\mathbf{F}\mathbf{F})^\dagger \cdot \rangle \mathbf{P}) - \gamma \mathbf{B}_{SE} \times \mathbf{P} \\ = & 2[(\Gamma_{bc} + \Gamma_{vdw,S})\langle \mathbf{S} \rangle + \Gamma_{vdw,F}\langle \mathbf{F} \rangle] - \left[\Gamma_{bc} + \Gamma_{vdw,S} + \Gamma_{vdw,F} \langle 2\mathbf{F} \cdot \mathbf{F} - \mathbf{F}\mathbf{F} \cdot - (\mathbf{F}\mathbf{F})^\dagger \cdot \rangle \right] \mathbf{P} \\ & - \gamma \mathbf{B}_{SE} \times \mathbf{P}. \end{aligned} \quad (S12)$$

With the presence of a continuous pump laser along the z axis, the spin polarization of the alkali-metal atoms can be approximated as a constant $\langle \mathbf{S} \rangle = S_0 \hat{z}$. In this case, the ^{87}Rb states that can be characterized by spin-temperature distribution, where $\langle F_z \rangle / \langle S_z \rangle = 2\langle \mathbf{F} \cdot \mathbf{F} - F_z^2 \rangle / q = q$. Assuming the effective field is negligible compared to the bias and feedback fields, the evolution equation for the nuclear spin polarization due to spin-exchange collisions reduces to

$$\frac{d}{dt}P_z = -\Gamma_{SE}P_z + R_{SE}, \quad (S13)$$

where $\Gamma_{SE} = \Gamma_{bc} + \Gamma_{vdw,S} + q\Gamma_{vdw,F}$. The first term $-\Gamma_{SE}P_z$ contributes to the relaxation, and $R_{SE} = 2\Gamma_{SE}S_0$ represents the noble-gas spin pumping originating from the spin-exchange interaction between alkali-metal and noble-gas spins.

(2) Feedback interaction. According to our measurement-feedback configuration, the average noble-gas spin polarization $\bar{P}_x$ is measured and subsequently fed back as a magnetic field denoted by $\mathbf{B}_f = -\alpha \bar{P}_x / \gamma \hat{y}$. Here α represents the feedback strength, which can be experimentally calibrated. Analogous to the externally applied magnetic field, the feedback magnetic field interacts with the nuclear spin via the Zeeman interaction. The specifics of this Zeeman interaction are elaborated upon in the following.

(3) Zeeman interaction. The Zeeman interaction causes the density operator of the noble-gas nuclear spin to evolve as follows

$$\frac{d}{dt}\rho = \frac{1}{i} [-\gamma \mathbf{K} \cdot (\mathbf{B}_z + \mathbf{B}_f), \rho]. \quad (S14)$$

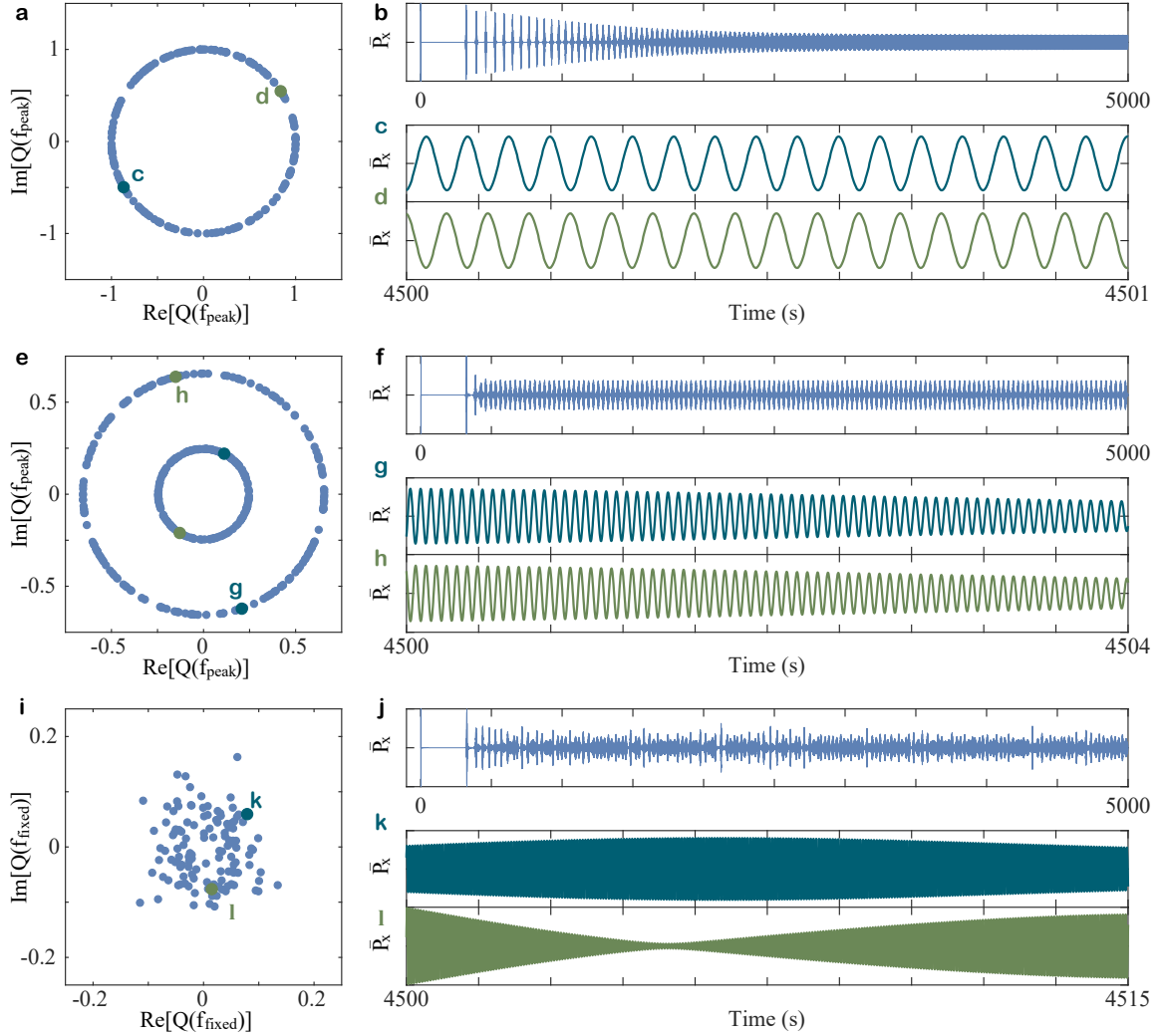
By multiplying both sides by $2\mathbf{K}$ and taking the trace, we derive the time evolution of the nuclear spin polarization

$$\frac{d}{dt}\mathbf{P} = \gamma \mathbf{P} \times (\mathbf{B}_z + \mathbf{B}_f). \quad (S15)$$

Substituting $\mathbf{B}_z = B_z(\mathbf{r})\hat{z}$ and $\mathbf{B}_f = -\alpha \bar{P}_x / \gamma \hat{y}$ into this equation yields the component evolution equations

$$\begin{cases} \dot{P}_x = \omega P_y + \alpha P_z \bar{P}_x \\ \dot{P}_y = -\omega P_x \\ \dot{P}_z = \alpha P_x \bar{P}_x \end{cases} \quad (S16)$$

where $\omega = \gamma B_z(\mathbf{r})$. Given that ω is a function of position $\mathbf{r}$, the polarization must also be treated as a function of position, or more meaningfully, as a function of the Larmor frequency ω . Thus, we express $P_i = P_i(\omega)$ for $i = x, y, z$. The averaged polarization



Supplementary Fig. 2 Phase distributions and corresponding time traces for limit cycle (a–d), quasi-periodic (e–h), and chaotic oscillations (i–l).

component $\bar{P}_x$ can be written as $\bar{P}_x = \int \rho(\omega') P_x(\omega') d\omega'$. Incorporating $P_i = P_i(\omega)$ and $\bar{P}_x$ into the component evolution equations allows us to derive the corresponding parts in Eq. (1) of the main text.

(4) Spin relaxation. The polarization of noble-gas nuclear spins can be disrupted or altered by various complex relaxation mechanisms. Notably, these include collisions with alkali-metal and other noble-gas atoms, spatial diffusion between regions of differing spin polarization, and spin relaxation during adsorption on vapor-cell wall. There are many works on these relaxation mechanisms, such as the comprehensive study by Ref. [5]. Ultimately, these mechanisms can be described semiclassically in terms of relaxation times T_1 and T_2 , which are incorporated into the evolution equations.

III. BREAKING OF CONTINUOUS TIME TRANSLATION SYMMETRY

In the main text, we demonstrate that the formation of the limit cycle and quasi-periodic oscillations is associated with spontaneous breaking of continuous time translation symmetry, as evidenced by their phase taking random values between 0 to 2π across repeated realizations. To gain further insights into the continuous time translation symmetry breaking, we consider the possibility that the spread over 2π in the time phase distribution originates from initial quantum fluctuations.

First, we consider the uncertainty of the initial state of our system. The uncertainty in the i -component ($i = x, y, z$) of the nuclear spin I is expressed as $\delta I_i = \sqrt{\langle I_i^2 \rangle - \langle I_i \rangle^2}$. For the system initiated near the equilibrium point of $\mathbf{P} = (0, 0, P_0)$, where $\langle I_x^2 \rangle = \langle I_y^2 \rangle = \langle I_z^2 \rangle = 1/4$, $\langle I_x \rangle = \langle I_y \rangle = 0$, and $\langle I_z \rangle = P_0/2$, the uncertainties are $\delta I_x = 1/2$, $\delta I_y = 1/2$, and $\delta I_z = \sqrt{1 - P_0^2}/2$.

The corresponding uncertainties in spin polarization are given by $\delta P_x = 2\delta I_x = 1$, $\delta P_y = 2\delta I_y = 1$, and $\delta P_z = 2\delta I_z = \sqrt{1 - P_0^2}$. For a system of N ^{129}Xe spins, the ensemble averaged uncertainty scales as $1/\sqrt{N}$, resulting in $\delta \bar{P}_x = 1/\sqrt{N}$, $\delta \bar{P}_y = 1/\sqrt{N}$, $\delta \bar{P}_z = \sqrt{(1 - P_0^2)/N}$. In our experiment, $N \approx 10^{18}$, $P_0 \approx 0.2$, we have the fluctuation $\delta \bar{P}_i > 10^{-9}$. Then, we include the fluctuation of $\delta \bar{P}_i = 10^{-9}$ in all components of the initial polarization into our model (Eq. 1 in the main text). We perform simulations for each case in Fig. 2g-i of the main text. For simulations we use $T_2 = 26$ s, $T_1 = 400$ s and $P_0 = T_1 R_{SE} = 0.2$. Below, we provide the results for three different dynamical phases.

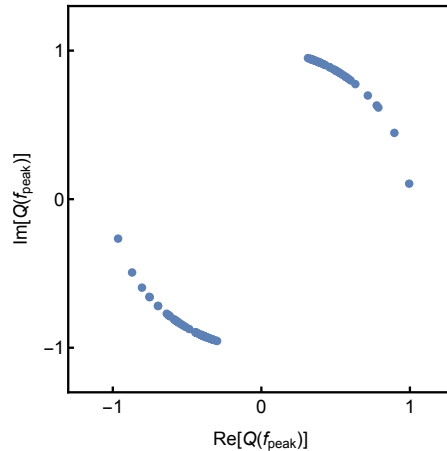
Limit cycle Figure 2a presents the simulated time phases of the limit cycle when only the initial quantum fluctuations are considered, displaying a random distribution covering $[0, 2\pi]$. Each point represents an individual simulation. Figure 2b illustrates the typical dynamical evolution of a single simulation, where oscillations spontaneously develop and then stabilize. We perform a Fourier transform of the spin polarization $\bar{P}_x$ during the steady-state regime and extract the complex Fourier amplitude at the peak frequency, $Q(f_{\text{peak}})$. To clearly illustrate how the phase of these oscillations varies from run to run, we select two specific $Q(f_{\text{peak}})$ points from Fig. 2a (labeled “c” and “d”) which have a time phase difference of almost π , and show their corresponding time traces $\bar{P}_x(t)$ in Figs. 2c and d, respectively.

Quasi-periodic oscillation Similarly, Figures 2e–h present the case of quasi-periodic oscillations. Unlike the limit cycle, the quasi-periodic oscillation features multiple spectral peaks. Several prominent frequency components are shown in Fig. 2e, and as a result, each individual simulation corresponds to multiple points in the plot. The dynamical evolution of the points in Fig. 2e is exemplified in Fig. 2f, where the signal exhibits wave packets with stable envelopes. We select two points labeled “g” and “h”, which exhibit a phase difference of almost π at the dominant frequency (the one with the highest peak), for detailed comparison. Since the phase difference between “g” and “h” is not apparent on the time scale of the full wave packet, we zoom in to examine the internal structure of a single packet. As shown in Fig. 2g and h, while the two signals exhibit similar amplitude modulation in envelopes, their underlying carrier oscillations are nearly in opposite phase at any given moment.

Chaotic phase For comparison, the distribution of simulated time phases in the chaotic regime is depicted in Fig. 2i, where all points scatter across a circular surface. This scattering arises from the lack of stable oscillations in chaos, resulting in significant variability in the Fourier amplitude at a fixed frequency across repeated simulations. The dynamical evolution of the points in Fig. 2i is exemplified in Fig. 2j, where the signal exhibits wave packets with varying envelopes. To illustrate the phase variability, we select two points labeled “k” and “l” which show a large phase difference at a fixed frequency, and display their respective time traces. At a specific moment, when the oscillation in Fig. 2k reaches a peak of a wave packet, the corresponding signal in Fig. 2l is near a zero crossing of a wave packet.

The numerical simulation agrees well with the experimental results. This suggests that the initial quantum fluctuation is sufficient to exhibit the breaking of continuous time translation symmetry. The spontaneous breaking of time translation symmetry indicates the emergence of continuous time crystals and quasi-crystals.

In contrast, the self-oscillations whose random phases are restricted to a subset of $[0, 2\pi]$ cannot be considered as CTCs. We have performed new numerical simulations to verify this point. In our case, for example, when parameters are set outside our experimental range — $\omega_c/2\pi = 0.012$ Hz with no magnetic field gradient ($g = 0$) and $\alpha = 1.92$ rad/s — 100 simulations initialized with fluctuation $\delta \bar{P}_i = 10^{-9}$ yield a phase distribution that fails to cover the full range of $[0, 2\pi]$, as shown in Fig. 3. This represents a trivial limit cycle rather than a CTC, differing from the result of a CTC in Fig. 2a, where the phase distribution is nearly uniform. Thus, the 2π random phase distribution arising from the spontaneous breaking of continuous time translation symmetry is a critical signature of a CTC.

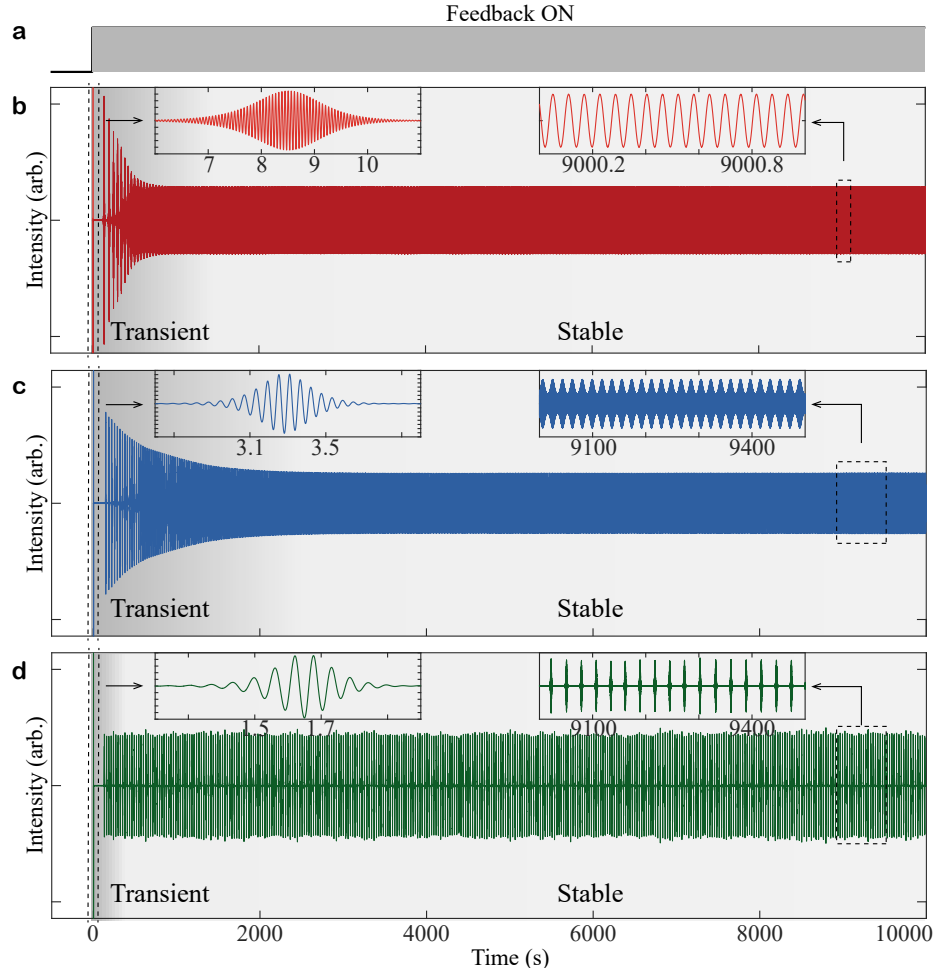


Supplementary Fig. 3 **Phase distribution of a trivial limit cycle (not a CTC).**

IV. PHASE DIAGRAM OVER A TWO-DIMENSIONAL PARAMETER SPAN

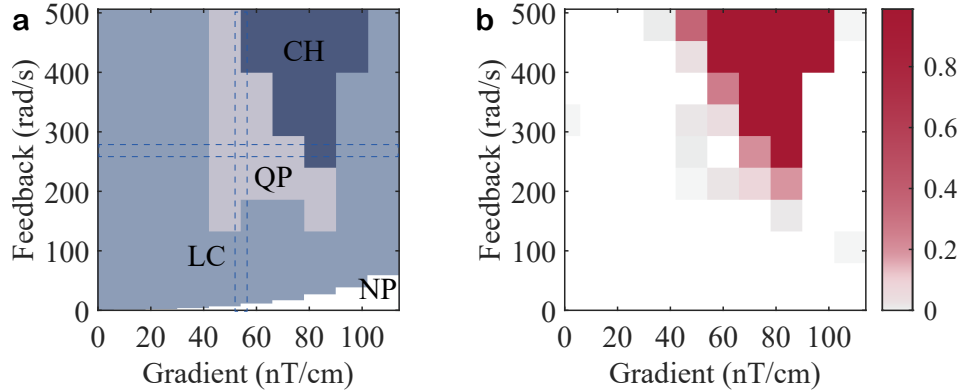
In order to more comprehensively demonstrate the dependence of the system's dynamical phase on the feedback strength α and the field gradient g , we systematically explore the (α, g) parameter space. Below, we detail the experimental procedure used to obtain this phase diagram and discuss the insights gained from it.

To construct the two-dimensional phase diagram, we maintain a constant magnetic-field gradient g while varying the resistance in series with the feedback-field coil to adjust the feedback strength α . This process is repeated for various g values to scan the parameter space. To thoroughly explore the parameter space, for each set of parameters, we begin by waiting approximately 2000 seconds with the feedback circuit disconnected to ensure that the ^{129}Xe spin polarization is fully aligned along the z axis. Once the feedback is activated, the system undergoes a transient phase before settling into stable oscillations, exhibiting distinct dynamic behaviors. Figure 4 illustrates the timing sequence of the feedback switch, along with representative time traces showing the establishment and stabilization of oscillations in the three dynamical phases. We measure the intensity changes of the probe beam, which is directly proportional to the average ^{129}Xe spin polarization, after the system dynamics have stabilized. For example, in the chaotic regime, data collection extends up to 10,000 seconds after the feedback is activated. To differentiate between chaotic and time-ordered dynamics (such as limit cycles and quasi-periodic behavior), we utilize the Chaos Decision Tree Algorithm. This method analyzes time series data and outputs a parameter K ; values approaching 1 indicate chaos, while values near 0 suggest periodic or quasi-periodic dynamics. Limit cycles and quasi-periodic behaviors can be intuitively distinguished by their spectral characteristics: quasi-periodicity is indicated by multiple peaks with incommensurate frequencies, whereas limit cycles are represented by a single peak, possibly with harmonics, though such harmonics were not observed in our experiments.



Supplementary Fig. 4 **Establishment of elementary oscillation pattern.** **a**, Feedback switch timing sequence. Representative time traces showing the transient build-up and stabilization of oscillations in the limit cycle (**b**), quasi-periodic (**c**), and chaotic (**d**) phases. The time traces presented in Fig. 2d-f of the main text are from an arbitrary 1000-s stationary time window.

The phase diagram is presented in Fig. 5a, and the K values, which are used to distinguish the dynamical phases, are calculated and shown in Fig. 5b. The experimental phase diagram spanned by (α, g) , showing distinct regimes: the no signal point, limit cycle, quasi-periodic orbits, and chaos. Compared to 1D parameter sweep of the previous version, the newly obtained 2D phase diagram offers a more comprehensive insight, in particular, we obtain some quantitative results. First, when the feedback strength is weak, ^{129}Xe spins exhibit a normal phase (no signal), with polarization aligned along z . As the feedback strength exceeds a critical threshold, the spin system enters a limit cycle phase. The threshold becomes higher as the magnetic field gradient increases. Second, only when the gradient exceeds a certain value (~ 40 nT/cm), phases beyond the limit cycle (such as quasi-periodic and chaotic dynamics) emerge, suggesting that the gradient plays a crucial role in the emergence of new dynamic phases. Third, after being supplemented by the 2D phase diagram, the conclusions drawn from the two phase transition curves in the previous version remains valid within a certain parameter range: over a broad gradient range (50 – 100 nT/cm), the system transitions from no signal to limit cycle, then to quasi-periodic phase, and finally to chaos as the feedback strength increases; over a broad feedback range ($\gtrsim 250$ rad/s), the system transitions from limit cycle to quasi-periodic phase, then to chaos, and eventually returns to limit cycle phase as the gradient increases. The portions enclosed by the dashed boxes in Fig. 5 correspond to the two phase transition curves of the previous version. Fourth, as the gradient increases, the quasi-periodic region becomes narrower, which may indicate an underlying competition mechanism within the system.



Supplementary Fig. 5 Experimental phase diagram spanned by (α, g) showing distinct regimes (a): the (NP) no signal point, (LC) limit cycle, (QP) quasi-periodic orbits, and (CH) chaos; and corresponding K value (b).

V. CRITERIA FOR DYNAMICAL BEHAVIOR

In this section, we provide a detailed description of the modified 0-1 test for chaos and an additional discriminating statistics for testing whether a system is chaotic – the maximum Lyapunov exponent. Furthermore, we present an analysis of the data in the experiment using these two methods, showing that the maximum Lyapunov exponent support the modified 0-1 test for chaos.

A. The modified 0-1 test for chaos

First, we describe the modified 0–1 test, which is central to the Chaos Decision Tree Algorithm [6] that we utilize to quantitatively identify chaos. For a given one-dimensional time-series of interest $\{\phi\}$, we can drive it to two-dimensional system with the following equations [6–8]:

$$\begin{cases} p_n = \sum_{j=1}^n \phi(j) \cos(jc) \\ q_n = \sum_{j=1}^n \phi(j) \sin(jc) \end{cases}, \quad (\text{S17})$$

where c is a random value bounded between 0 and 2π . The above transformation can be traced back to the components of the solution to the skew-product system:

$$\begin{cases} \dot{\theta} = c + \phi \\ \dot{p} = \phi \cos \theta \\ \dot{q} = \phi \sin \theta \end{cases}. \quad (\text{S18})$$

It has been shown that the dynamics is bounded if the underlying dynamics is non-chaotic and unbounded if the underlying dynamics is chaotic [9]. Moreover, the p and q components behave like a Brownian motion if the underlying dynamics is chaotic [10]. In order to use this principle to determine whether the underlying dynamics are regular or chaotic, the mean square displacement is defined:

$$M(n) = \frac{1}{N} \sum_{j=1}^N \{[p(j+n) - p(j)]^2 + [q(j+n) - q(j)]^2\}. \quad (\text{S19})$$

If the behaviour is Brownian, i.e. the underlying dynamics is chaotic, then $M(n)$ grows linearly in time; if the behaviour is bounded, i.e. the underlying dynamics is regular (periodic or quasi-periodic), then $M(n)$ is bounded. The above description is quantified by K , which is defined as the growth rate of $M(n)$. The value of K approaches 1 signifies chaotic systems, whereas the value nearing 0 indicates periodic or quasi-periodic systems.

B. Maximum Lyapunov exponent

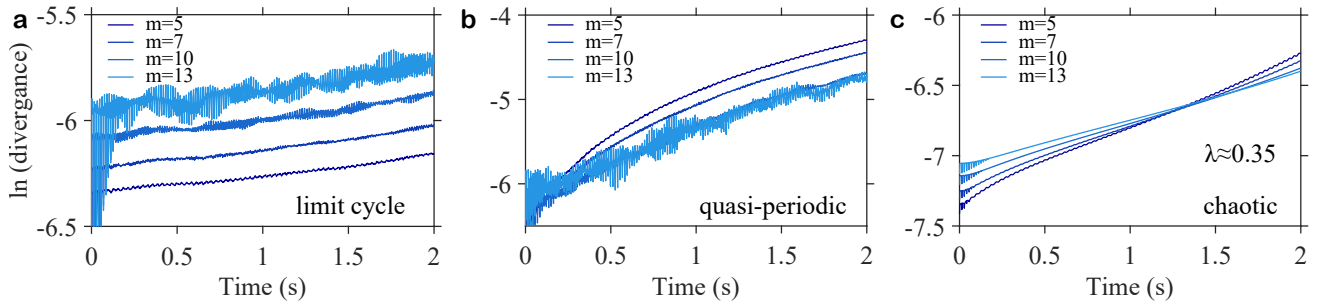
Chaos arises from the exponential growth of infinitesimal perturbations. This exponential instability is characterized by the spectrum of Lyapunov exponents. A positive largest Lyapunov exponent indicates chaos: if $\lambda > 0$, then nearby trajectories separate exponentially, and if $\lambda < 0$, then nearby trajectories stay close to each other.

Since a scalar sequence in itself does not properly represent the multi-dimensional phase space of the dynamical system, phase-space reconstruction is required to compute the maximum Lyapunov exponent of a time series. The most important phase space reconstruction technique is the method of delays. For a time series $\{s_n\}$, vectors in a new space, the embedding space, are formed from time delayed values of the scalar measurements:

$$s_n = (s_{n-(m-1)\tau}, s_{n-(m-2)\tau}, \dots, s_n),$$

where the number m of elements is the embedding dimension, and the time τ is referred to as the delay. Consider the representation of the time series data as a trajectory in the embedding space, and assume that we observe a very close return $s_{n'}$ to a previously visited point s_n . Then one can consider the distance $\Delta_0 = s_n - s_{n'}$ as a small perturbation, which should grow exponentially in time. Its future can be read from the time series $\Delta_l = s_{n+l} - s_{n'+l}$. If one finds that $\Delta_l \sim \Delta_0 e^{\lambda l}$, then λ is the maximum Lyapunov exponent. In the following, we calculate the maximum Lyapunov exponent of our experimental data using the TISEAN software package [11], and compare the result with the Chaos Decision Tree Algorithm. The embedding time delay $\tau = 14$ ms is used for all the three sets of data in Fig. 6a-c.

Figure 6a illustrates the analysis of the maximum Lyapunov exponent for varying embedding dimensions from the data in the limit cycle phase presented in the main text, Fig. 2. The absence of linearly scaled regions for the rising embedding dimension m suggests non-chaotic behavior, consistent with the K value approaching 0. The same holds true in Figure 6b. Figure 6c presents the analysis of data in the chaotic phase, which exhibits a clear range of linear increase with an identical slope for the rising embedding dimension m , indicating a positive Lyapunov exponent. Correspondingly, the K value nears 1. The results of three different sets of data all indicate that the analysis of the K value and the maximum Lyapunov exponent support each other in identifying chaotic behavior.



Supplementary Fig. 6 Time series analysis of experimental data for limit cycle (a), quasi-periodic (b), and chaotic (c) phases.

C. Correlation dimension

The correlation dimension D_2 quantifies the fractal characteristics of a dynamical system's attractor by describing how the number of point pairs within a given distance ϵ , denoted $C_2(\epsilon)$, scales with ϵ . For example, a simple limit cycle traces out

a one-dimensional closed curve in phase space. At a given point on the limit cycle, the number of points within a spherical neighborhood of radius ε scales as $C_2(\varepsilon) \sim \varepsilon^1$ as ε increases, yielding a correlation dimension $D_2 \approx 1$. For quasi-periodic orbits evolving on an N -dimensional torus, the points uniformly fill an N -dimensional manifold, resulting in $D_2 \approx N$. In contrast, chaotic attractors typically exhibit non-integer correlation dimensions, reflecting their underlying fractal geometry and complex, self-similar structure in phase space. Given a reconstructed phase space ($\{s_n\}$), the correlation dimension is estimated via the correlation sum:

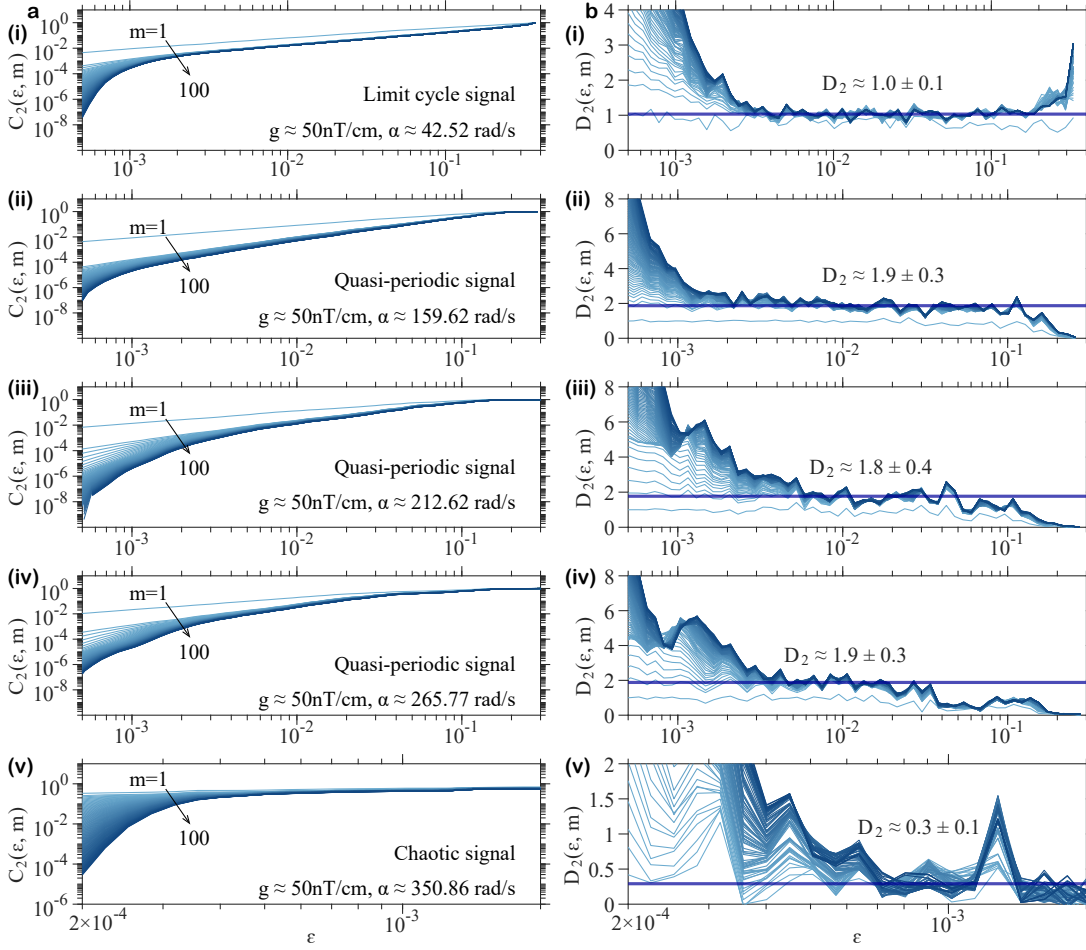
$$C_2(m, \varepsilon) = \frac{1}{N_{\text{pairs}}} \sum_{j=mk}^N \sum_{j-w} \Theta(\varepsilon - |s_j - s_k|), \quad (\text{S20})$$

where N_{pairs} is the number of pairs of points covered by the sums, $\Theta(\cdot)$ is the Heaviside step function that counts the number of pairs (s_j, s_k) separated by less than ε , and w is the Theiler window that excludes temporally correlated points from the pair. On sufficiently small length scales and when the embedding dimension m exceeds the correlation dimension of the attractor,

$$C_2(m, \varepsilon) \sim \varepsilon^{D_2}, \quad (\text{S21})$$

such that a log-log plot of $C_2(m, \varepsilon)$ versus ε yields a linear region whose slope provides an estimate of the correlation dimension D_2 . Since the correlation dimension is not known a priori, we check for convergence of the estimated values of D_2 in m . We employ the TISEAN package [11] to compute the correlation dimension of our data. Below, we present our calculation results and analysis for the data corresponding to the three dynamical phases.

Figure 7 shows the calculation results of the correlation dimension of our data. The first row (i) in Fig. 7 presents the analysis of the limit cycle data (main text, Fig. 2d). Panel a displays the log-log plot of the correlation sum C_2 versus the radius ε across



Supplementary Fig. 7 Calculation of correlation dimension. **a**, Log-log plot of correlation sum C_2 versus threshold radius ε for an increasing number of embedding dimensions m . **b**, Slope from panel **a** as function of $\log(\varepsilon)$, which converges to a constant value for a range of ε , shown by the dark blue line. The first row (i) presents the analysis of the limit cycle data. The rows (ii)-(iv) present the analysis of the quasi-periodic data. The last row (v) presents the analysis of the chaotic data.

increasing embedding dimensions m . Panel b depicts the slope of the curves from Fig. 7a, corresponding to the correlation dimension D_2 . The linear range of converged curves is observed around the ϵ range of 0.003 to 0.16, yielding $D_2 \approx 1.0 \pm 0.1$, indicated by the dark blue line. The integer value of D_2 reflects the periodic nature of the time series, consistent with the previously calculated $K \approx 0.0016$ and the absence of a positive Lyapunov exponent.

The second row (ii) presents the analysis of the quasi-periodic data (main text, Fig. 2e). Using the same procedure, the computed correlation dimension is close to an integer value of 2 within the margin of error. Considering the diversity of the quasi-periodic phase, such as the varying number of spectral peaks under different experimental parameters, we analyze additional quasi-periodic data under various conditions, shown in rows (iii)–(iv). In all cases presented, the resulting correlation dimension are close to an integer value of 2.

The last row (v) shows the analysis of the chaotic data (main text, Fig. 2f). Applying the same procedure, we obtain a non-integer correlation dimension of approximately $D_2 \approx 0.3 \pm 0.1$. This result further corroborates the chaotic nature, as indicated by the previously calculated $K \approx 0.9819$ and the Lyapunov exponent $\lambda \approx 0.35$.

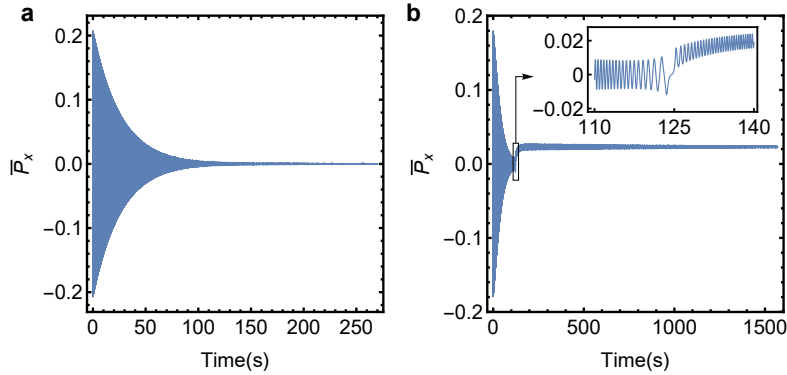
VI. ANALYSIS OF PARAMETER DEPENDENCE

In this section, we provide a brief discussion on the influence of key experimental parameters, including gas pressure, pump and probe power, and the direction of the feedback field.

Gas density dependence According to the spin-exchange mechanisms, both the binary collision rate and the three-body formation rate per Xe atom scale proportionally with the Rb number density $[\text{Rb}]$. Hence, the Xe spin-exchange optical pumping rate is directly proportional to $[\text{Rb}]$. The Xe number density $[\text{Xe}]$ primarily affects the spin-exchange interaction by participating as the third body in the van der Waals molecule formation process. On one hand, a higher $[\text{Xe}]$ enhances the formation rate of van der Waals molecules; on the other hand, it reduces their lifetime τ_{vdw} . Typically, the negative impact of latter outweighs the benefits of the former. From the perspective of spin-exchange rate, a higher Rb density and a lower Xe density seems favorable. However, increasing the Rb density also accelerates Xe spin relaxation due to collisions, and an insufficient Xe density undermines signal readout sensitivity. Therefore, the Rb number density should not be too high, nor should the Xe number density be too low. Determining the optimal Xe-Rb fractional ratio theoretically is complex. In this experiment, based on experience, we have chosen a ^{129}Xe gas pressure of 50 Torr, with the ^{87}Rb saturated vapor temperature set to 105°C.

Power Dependence The power of the pump and probe beams would influence the oscillation behavior. For example, altering the pump beam power changes the absorption rate of Rb atoms for photons, thereby affecting the spin polarization of ^{87}Rb . This influences both the polarimetric sensitivity that impacts the feedback strength α , and the spin exchange between Rb and Xe. Both factors influence the oscillation behavior of ^{129}Xe . Similarly, adjusting the probe beam power also affects polarimetric sensitivity, subsequently impacting the feedback strength α , and ultimately influencing the oscillation behavior of ^{129}Xe .

Alternative feedback configurations In the main text, the feedback field is applied along the y-axis, given by $B_y = -\alpha \bar{P}_x / \gamma$. Furthermore, we perform new simulations with the feedback field applied along the x-axis, i.e., $B_x = -\alpha \bar{P}_x / \gamma$. These simulations reveal that the system evolves toward a steady state, characterized either by vanishing transverse polarization or by a non-zero constant transverse polarization. Two representative examples are shown in Fig. 8: panel a illustrates a case where the system relaxes to a state with $\bar{P}_x = 0$, while panel b shows the system stabilizing at a finite, constant $\bar{P}_x$.



Supplementary Fig. 8 **Representative dynamics with the feedback field applied along the x-axis.** The parameters are chosen as follows: $\alpha = 76.92$ rad/s, $g = 3.89$ nT/cm for **a**; and $\alpha = -769.23$ rad/s, $g = 3.89$ nT/cm for **b**.

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